

Charles Langer

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Education

University of Utah, Salt Lake City

December 2026

Bachelor of Science in Electrical Engineering, GPA: 3.9

- Six-Time Dean's List

Courses Completed

- Classical Control Systems
- Linear Active Circuit Design
- Cyber-Physical Systems
- Power System Analysis
- Introduction to Power
- Object-Oriented Programming
- Fundamental Electromagnetics
- PDEs for Engineers
- Signals and Systems

Affiliations

- Institute of Transportation Engineers Student Chapter Member
 - Vice-President of University of Utah Badminton Club
- July 2025 – Present
May 2024 – Present

Technical Projects

Traction Power Substation Design

Spring 2026

- Designed and reviewed single-line diagrams for AC utility interconnection and 750 V DC traction power systems, including Delta-Wye transformers and rectifier conversion
- Interpreted electrical drawings, metering schemes, and IEEE protective relay device numbers to ensure proper system coordination and code compliance
- Developed TPSS equipment layout and duct bank/conduit schematics, applying protection principles to enhance feeder, transformer, and rectifier reliability

Modeling & Controlling of a Flexible System

Fall 2025

- Worked alongside a team to model a flexible crane-like system in MATLAB using Bode plots and dynamic analysis
- Designed and tuned a PID controller using root locus, phase margin, steady-state error analysis to meet settling time and overshoot specifications
- Facilitated project coordination and consolidated modeling and control results into a formal technical report

Night Light Circuit

Spring 2025

- Designed, and implemented a light-activated control system that processed sensor input and executed automatic on/off switching logic based on ambient light conditions.
- Translated control logic into hardware and validated switching behaviors using LTspice and electrical measurements.

Work Experience

Whole Foods

September 2022 – June 2023

Prepared Foods Team Member

Salt Lake City, UT

- Maintained quality control standards by monitoring inventory quality, supported end-of-day closeout procedures, and ensured adherence to established health, and safety standards.
- Demonstrated reliability and attention to detail by following structured procedures, executing assigned tasks accurately, and earning expanded responsibilities beyond a part-time scope.

City of Burbank, CREST Program

October 2019 – March 2020

Youth Services Worker

Burbank, CA

- Worked independently to record, organize, and archive confidential administrative documents to City's Attorney Office
- Helped coordinate delivery of documents and files between Courthouse, Police Department and City Hall.

Technical Skills

- Adept in MATLAB, Java, and Python based programming.
- Familiar with NFPA 70-NEC, IEEE, NFPA-130 and technical design software programs such as AutoCAD
- Proficient with Microsoft Word, PowerPoint, and Excel